

## Unit V Cluster Analysis

**Types of Data in Cluster Analysis – A Categorization of Major Clustering Methods – Partitioning Methods – Hierarchical methods – Density-Based Methods – Grid-Based Methods – Model-Based Clustering Methods – Clustering High-Dimensional Data – Constraint-Based Cluster Analysis – Outlier Analysis.**

### Cluster Analysis

Cluster is a group of objects that belongs to the same class. In other words, similar objects are grouped in one cluster and dissimilar objects are grouped in another cluster.

### What is Clustering?

Clustering is the process of making a group of abstract objects into classes of similar objects.

### Points to Remember

- A cluster of data objects can be treated as one group.
- While doing cluster analysis, we first partition the set of data into groups based on data similarity and then assign the labels to the groups.
- The main advantage of clustering over classification is that, it is adaptable to changes and helps single out useful features that distinguish different groups.

### Cluster Analysis Examples

Some cluster analysis examples are given below:

1. **Markets-** Cluster analysis helps marketers to find different groups in their customer bases and then use the information to introduce targeted marketing programs.
2. **Land** - It is used to identify areas of the same land used in an earth observation database.
3. **Insurance** - Cluster analysis helps to identify groups who hold a motor insurance policy with a high average claim cost.
4. **Earthquake Studies** - Cluster analysis helps to observe earthquakes.
5. **City-Planning** - Cluster analysis helps to recognize houses on the basis of their types, house value and geographical location.

### Examples of Cluster Analysis in Real Life

**Cluster analysis** is a technique used in machine learning that attempts to find clusters of observations within a dataset.

The goal of cluster analysis is to find clusters such that the observations within each cluster are quite similar to each other, while observations in different clusters are quite different from each other.

The following examples show how cluster analysis is used in various real-life situations.

### **Example 1: Retail Marketing**

Retail companies often use clustering to identify groups of households that are similar to each other.

For example, a retail company may collect the following information on households:

- Household income
- Household size
- Head of household Occupation
- Distance from nearest urban area

They can then feed these variables into a clustering algorithm to perhaps identify the following clusters:

- Cluster 1: Small family, high spenders
- Cluster 2: Larger family, high spenders
- Cluster 3: Small family, low spenders
- Cluster 4: Large family, low spenders

The company can then send personalized advertisements or sales letters to each household based on how likely they are to respond to specific types of advertisements.

### **Example 2: Streaming Services**

Streaming services often use clustering analysis to identify viewers who have similar behavior.

For example, a streaming service may collect the following data about individuals:

- Minutes watched per day
- Total viewing sessions per week
- Number of unique shows viewed per month

Using these metrics, a streaming service can perform cluster analysis to identify high usage and low usage users so that they can know who they should spend most of their advertising dollars on.

### **Example 3: Sports Science**

Data scientists for sports teams often use clustering to identify players that are similar to each other.

For example, professional basketball teams may collect the following information about players:

- Points per game
- Rebounds per game
- Assists per game
- Steals per game

They can then feed these variables into a clustering algorithm to identify players that are similar to each other so that they can have these players practice with each other and perform specific drills based on their strengths and weaknesses.

#### **Example 4: Email Marketing**

Many businesses use cluster analysis to identify consumers who are similar to each other so they can tailor their emails sent to consumers in such a way that maximizes their revenue.

For example, a business may collect the following information about consumers:

- Percentage of emails opened
- Number of clicks per email
- Time spent viewing email

Using these metrics, a business can perform cluster analysis to identify consumers who use email in similar ways and tailor the types of emails and frequency of emails they send to different clusters of customers.

#### **Example 5: Health Insurance**

Actuaries at health insurance companies often used cluster analysis to identify “clusters” of consumers that use their health insurance in specific ways.

For example, an actuary may collect the following information about households:

- Total number of doctor visits per year
- Total household size
- Total number of chronic conditions per household
- Average age of household members

An actuary can then feed these variables into a clustering algorithm to identify households that are similar. The health insurance company can then set monthly premiums based on how often they expect households in specific clusters to use their insurance.

#### **Applications of Cluster Analysis**

- Clustering analysis is broadly used in many applications such as market research, pattern recognition, data analysis, and image processing.
- Clustering can also help marketers discover distinct groups in their customer base. And they can characterize their customer groups based on the purchasing patterns.
- In the field of biology, it can be used to derive plant and animal taxonomies, categorize genes with similar functionalities and gain insight into structures inherent to populations.
- Clustering also helps in identification of areas of similar land use in an earth observation database. It also helps in the identification of groups of houses in a city according to house type, value, and geographic location.
- Clustering also helps in classifying documents on the web for information discovery.
- Clustering is also used in outlier detection applications such as detection of credit card fraud.

- As a data mining function, cluster analysis serves as a tool to gain insight into the distribution of data to observe characteristics of each cluster.

### Requirements of Clustering in Data Mining

The following points throw light on why clustering is required in data mining –

- **Scalability** – We need highly scalable clustering algorithms to deal with large databases.
- **Ability to deal with different kinds of attributes** – Algorithms should be capable to be applied on any kind of data such as interval-based (numerical) data, categorical, and binary data.
- **Discovery of clusters with attribute shape** – The clustering algorithm should be capable of detecting clusters of arbitrary shape. They should not be bounded to only distance measures that tend to find spherical cluster of small sizes.
- **High dimensionality** – The clustering algorithm should not only be able to handle low-dimensional data but also the high dimensional space.
- **Ability to deal with noisy data** – Databases contain noisy, missing or erroneous data. Some algorithms are sensitive to such data and may lead to poor quality clusters.
- **Interpretability** – The clustering results should be interpretable, comprehensible, and usable.

### Clustering Methods

Clustering methods can be classified into the following categories –

- Partitioning Method
- Hierarchical Method
- Density-based Method
- Grid-Based Method
- Model-Based Method
- Constraint-based Method

### Partitioning Method

Suppose we are given a database of ‘n’ objects and the partitioning method constructs ‘k’ partition of data. Each partition will represent a cluster and  $k \leq n$ . It means that it will classify the data into k groups, which satisfy the following requirements –

- Each group contains at least one object.
- Each object must belong to exactly one group.

### Points to remember –

- For a given number of partitions (say k), the partitioning method will create an initial partitioning.

- Then it uses the iterative relocation technique to improve the partitioning by moving objects from one group to other.

### **Hierarchical Methods**

This method creates a hierarchical decomposition of the given set of data objects. We can classify hierarchical methods on the basis of how the hierarchical decomposition is formed. There are two approaches here –

- Agglomerative Approach
- Divisive Approach

#### **Agglomerative Approach**

This approach is also known as the bottom-up approach. In this, we start with each object forming a separate group. It keeps on merging the objects or groups that are close to one another. It keep on doing so until all of the groups are merged into one or until the termination condition holds.

#### **Divisive Approach**

This approach is also known as the top-down approach. In this, we start with all of the objects in the same cluster. In the continuous iteration, a cluster is split up into smaller clusters. It is down until each object in one cluster or the termination condition holds. This method is rigid, i.e., once a merging or splitting is done, it can never be undone.

### **Approaches to Improve Quality of Hierarchical Clustering**

Here are the two approaches that are used to improve the quality of hierarchical clustering –

- Perform careful analysis of object linkages at each hierarchical partitioning.
- Integrate hierarchical agglomeration by first using a hierarchical agglomerative algorithm to group objects into micro-clusters, and then performing macro-clustering on the micro-clusters.

### **Density-based Method**

This method is based on the notion of density. The basic idea is to continue growing the given cluster as long as the density in the neighborhood exceeds some threshold, i.e., for each data point within a given cluster, the radius of a given cluster has to contain at least a minimum number of points.

#### **Grid-based Method**

In this, the objects together form a grid. The object space is quantized into finite number of cells that form a grid structure.

### **Advantages**

- The major advantage of this method is fast processing time.
- It is dependent only on the number of cells in each dimension in the quantized space.

### **Model-based methods**

In this method, a model is hypothesized for each cluster to find the best fit of data for a given model. This method locates the clusters by clustering the density function. It reflects spatial distribution of the data points.

This method also provides a way to automatically determine the number of clusters based on standard statistics, taking outlier or noise into account. It therefore yields robust clustering methods.

### **Constraint-based Method**

In this method, the clustering is performed by the incorporation of user or application-oriented constraints. A constraint refers to the user expectation or the properties of desired clustering results. Constraints provide us with an interactive way of communication with the clustering process. Constraints can be specified by the user or the application requirement.

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